1.List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

IP Class	Address Range	Example
Class A	1.0.0.0 – 126.255.255.255	Large company network
Class B	128.0.0.0 – 191.255.255.255	University/college network
Class C	192.0.0.0 – 223.255.255.255	Home Wi-Fi router
Class D	224.0.0.0 – 239.255.255.255	Video streaming/multicast
Class E	240.0.0.0 – 255.255.255.255	Experimental/research use

2.Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses. show your working steps for each calculation.

Given:

- **IP Address: 192.168.10.45**
- **Subnet Mask: 255.255.255.0 (/24)**

Step 1: Network Address

With /24, the first 3 octets are the network part.

Network Address = 192.168.10.0

Step 2: Broadcast Address

The last octet can go from 0 to 255.

Broadcast Address = 192.168.10.255

Step 3: Valid Host Range

- First usable = Network + 1 → **192.168.10.1**
- Last usable = Broadcast − 1 → **192.168.10.254**

Item	Address
Network Address	192.168.10.0
First Host	192.168.10.1
Last Host	192.168.10.254
Broadcast Address	192.168.10.255
Valid Hosts	**254**

3.Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

- **Using 10.0.0.0/24 and needing 4 separate subnets, we can divide it into 4 equal /26 subnets.**

Group	Network Address	Subnet Mask	Usable Host Range
Delivery Partners	10.0.0.0	255.255.255.192 (/26)	10.0.0.1 – 10.0.0.62
Restaurants	10.0.0.64	255.255.255.192 (/26)	10.0.0.65 – 10.0.0.126
Customers	10.0.0.128	255.255.255.192 (/26)	10.0.0.129 – 10.0.0.190
Admins	10.0.0.192	255.255.255.192 (/26)	10.0.0.193 – 10.0.0.254

4.You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet. Constraint: Show your calculation steps and final answers.

Given:

- **Network: 172.16.0.0/16**
- **Required subnets: At least 20**

➢ **Step 1: Find borrowed bits**

We need:

$2^n \geq 20$

- $2^4 = 16$
- $2^5 = 32$

So, 5 bits are borrowed.

➢ **Step 2: New subnet mask**

Original: **/16**
Borrowed: **5 bits**

New prefix = 16 + 5 = /21

Subnet mask:

255.255.248.0

- **Step 3: Hosts per subnet**

Remaining host bits:

32 − 21 = 11 bits

Total addresses:

2^{11} = 2048

Usable hosts:

2048 − 2 = 2046

Item	**Answer**
Original Network	172.16.0.0/16
Required Subnets	At least 20
Bits Borrowed	**5 bits**
Number of Subnets	**32**
New Subnet Mask	**255.255.248.0 (/21)**
Usable Hosts/Subnet	**2046**

5.A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

- The IP address **192.168.1.300** is invalid because each part of an IPv4 address must be between **0 and 255**. The value **300** is outside this range.

For the **192.168.1.0/24** subnet, a valid host IP could be **192.168.1.100** or **192.168.1.200**.

So, the device should be given a valid IP address where the last number is between **1 and 254**.